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ORIGINAL RESEARCH

An intelligent sliding mode control for stabilization of parallel converters feeding CPLs in DC-microgrid

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Abstract

The constant power loads (CPLs) threaten the stability of the DC microgrids (DCMGs) due to the negative impedance characteristic, which leads to voltage oscillations or a blackout in the DCMG. This paper proposes a new droop control for paralleled DC–DC converters feeding CPLs in the DCMG to improve the stability of the whole system. In this approach, an optimization droop gain is conducted for each converter to guarantee current sharing among paralleled converters. Furthermore, an intelligent single input interval type-II fuzzy logic controller (SIT2-FLC) is developed in a model-free framework to further ameliorate the current sharing and voltage regulation in the DCMG when supplying the CPLs. An extended-state observer (ESO) is adopted in the model-free controller to estimate the unmodelled dynamics in the DC-DC converters feeding CPLs. Since an estimation observer error often appears in the presence of uncertainties, a sliding mode control (SMC) technique is proposed to remove the estimation error and enhance the system stability. The studied DCMG is developed in hardware-in-the-loop (HIL) with various levels of CPL's power to verify the applicability of the suggested mode-free intelligent controller.

1 | INTRODUCTION

Recently, the DC microgrid (DCMG) has gained more and more attention due to the highlighted advantages over the AC microgrid such as simple control, absence of harmonic and synchronization, reactive power control issues, and high efficiency, and easy compatibility between sources and loads. Moreover, with growth in the utilization of renewable energy sources (RESs) and energy storage systems (ESSs) in the DCMGs with the integration of DC–DC converters led to a more developed DCMG concept in several applications such as more electric aircraft, all-electric ships, hybrid electric vehicles etc. [1].

In the DCMG configuration, the distributed generations (DGs) are connected to the DC bus via a DC–DC converter with different line resistance, which is loaded by various loads. In addition, in the islanded DCMG, power electronic converters are paralleled to inject the power into the loads, which is improved the reliability of the system. However, utilizing paralleled converters with various line resistance in the DCMG have a challenging problem due to the current sharing among them. To overcome this problem in the stand-alone DCMG, several studies have introduced control strategies [2]. They can be classified into linear and non-linear droop control methods. In ref. [3], a linear method based on the droop control is proposed to increase damping and eliminate the current sharing error among paralleled power converters in the DCMG. In ref. [4, 5], nonlinear droop control is employed to improve the current sharing in the DCMG. To achieve the state of charge (SOC) balancing, adaptive droop control is developed in ref. [6]. These methods suffer from poor current sharing when the line resistances of the converters are different.

On the other hand, power electronic converters when tightly regulated, act as a constant power load (CPL). CPLs in the DCMG have negative impedance characteristics, which may destabilize the DC bus in the DCMG and cause voltage oscillation and consequently blackout in the whole islanded DCMG [7, 8]. Furthermore, adding a capacitor or resistor known as the passive damping method can improve the stability of the DCMG, whereas they are costly and increase the size of the converters [9]. Active damping strategies stabilize the DCMG feeding CPLs utilizing virtual resistor, virtual capacitor, and virtual impedance to improve the stability of the

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DCMG by modifying the control loops [10]. These methods increase damping by injecting the power into the system. However, the approaches based on the small-signal models such as active damping methods can only guarantee small-signal stability around the operating point. Therefore, these control methods have a problem from the stability point of view when a large disturbance happens in the DCMGs and the system may be unstable. Moreover, the aforementioned works consider one DC–DC power converter in the DCMG, and the effect of paralleling converters is eliminated. Due to the non-linear nature of CPLs in the stand-alone DCMG, non-linear control approaches in the large signal sense should be developed for the stabilization of the DCMG. Several non-linear control techniques have been introduced to mitigate the negative effect of CPLs in the DCMG. In ref. [11], a passivity-based control approach is presented to control a boost converter feeding ideal CPLs. This technique stabilizes the converter but the system's accurate model is required, which imposes complexity on the controller. In ref. [12], an SMC controller is suggested to control the duty ratio of the switch in the DC–DC converter loaded by CPLs. This method stabilizes the system over a wide range of loads. However, in this method the cost of implementation of the system is increased since the measuring of the capacitor current is needed. A model predictive control approach is suggested in ref. [13]. Model predictive control (MPC) suffers from a real-time computational burden, which degrades the applicability of this controller. Moreover, several controllers based on the fuzzy system such as robust non-fragile fuzzy control [14], distributed fuzzy polynomial controller [15], distributed sampled-data fuzzy controller [16, 17], state feedback controller [18], intelligent fuzzy controller [8, 19–21] have been investigated in the literature. Reviewing the previous works reveals that, in the existing non-linear and linear control methods, the transient performance of the DCMG is not considered. Moreover, current sharing issues in the paralleled DC–DC converters with different line resistance degrade the performance of the DCMGs, which is left behind in the above references. Despite the rich literature on DCMG with CPLs, almost all research works developed their control algorithms on the power electronic systems without any inaccuracy, and the behaviour of CPL is ideal. However, in practice applications, the power converts in the DCMG are faced with many uncertainties and time-varying CPLs which cannot be addressed by the conventional deterministic control methodologies. Thus, besides the stability problems in the DCMGs with CPLs, further efforts are required to provide better current sharing of paralleled DC–DC converters in the DCMGs with CPLs.

With the aim of addressing the deployment challenges of CPLs, this paper proposes a new droop control and a non-linear controller for current sharing and voltage stabilization in the DCMGs. The contributions of the current work are as:

1. A non-linear droop control method is developed to ameliorate the current sharing among paralleled DC–DC converters with non-equal line resistance in the islanded DCMG.

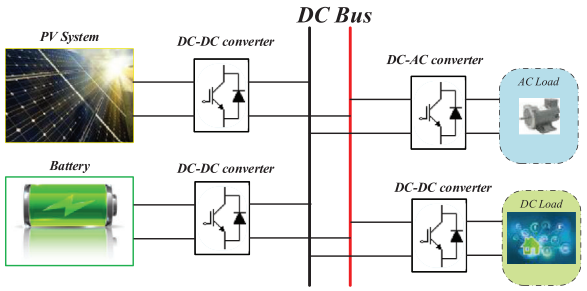


FIGURE 1 Illustration of a typical DCMG

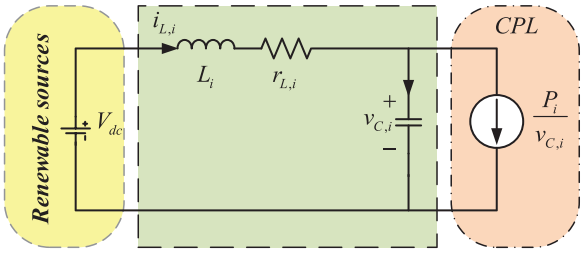


FIGURE 2 Modelling of CPL subsystem in the DCMG

2. The proposed droop controller finds the optimum value of the droop gain in each converter based on the line resistance, which is effectively improved the current sharing among paralleled converters.
3. Furthermore, a novel model-free sliding mode control (SMC) approach with an intelligent single input interval type-II fuzzy logic controller (ISIT2-FLC) is suggested to guarantee the stability of the paralleled converters with CPLs in the DCMG. In the proposed control scheme, an extended state observer (ESO) is implemented to eliminate the effects of unmodelled dynamics of the CPLs in the DCMG framework.
4. The suitability of the proposed controller has been experimentally tested by Hardware-in-the-loop (HIL) simulation under different levels of loading conditions.

The paper is organized as follows. The overall structure of the DCMG is modelled in Section 2. In Section 3, the proposed non-linear droop control and the current sharing issue is discussed. Section 4 presents the new ISIT2-FLC method and proof of the stability of the control approach is described. To validate the efficacy of the suggested controller, experimental results are presented in Section 5. Section 6 concludes the paper.

2 | MODELING OF THE DC MICROGRID

A typical DCMG configuration is shown in Figure 1, which is used in several applications such as smart homes, All-electric ships, and electric aircraft. The considered DCMG contains several paralleled converters feeding CPLs, RESs, and ESSs. Figure 2 illustrates the circuit of a CPL subsystem in the DCMG framework.

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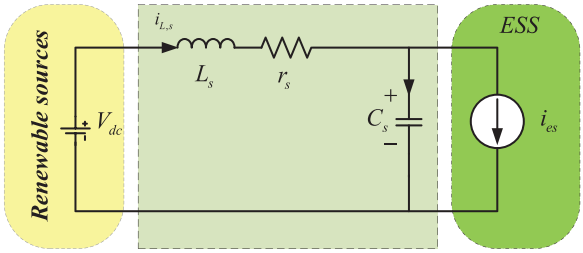


FIGURE 3 Modelling of the source and ESS

In Figure 2, $i_{L,i}$ and $v_{c,i}$ are the inductance current and capacitor voltage, respectively. V_{dc} is the input voltage, L_i is the inductance of the converter and $r_{L,i}$ is the internal resistance of the inductance. As shown in Figure 2, to drive the dynamical model of the DCMG, first, one DC source is considered, which is loaded by one CPL. The dynamic model of the ith CPL subsystem in the DCMG is obtained as follows [22]:

$$\begin{cases} \dot{i}_{L,i} = -\frac{r_{L,i}}{L_i} i_{L,i} - \frac{1}{L_i} v_{C,i} + \frac{1}{L_i} V_{dc} \\ \dot{v}_{C,i} = \frac{1}{C_i} i_{L,i} - \frac{1}{C_i} \frac{P_i}{v_{C,i}} \end{cases} \tag{1}$$

where P_i is the power of the ith CPL. Moreover, C_i is the output capacitance. It should be noted that the ideal CPL is demonstrated as a current sink. The dynamic model of the DC–DC converter with ESS is shown in Figure 3.

As depicted in Figure 3, the RESs are considered as the sources and the dynamic model of this subsystem loaded with ESSs can be written as:

$$\begin{cases} \dot{i}_{L,s} = -\frac{r_s}{L_s} i_{L,s} - \frac{1}{L_s} v_{C,s} + \frac{1}{L_s} V_{dc} \\ \dot{v}_{C,s} = \frac{1}{C_s} i_{L,s} - \frac{1}{C_s} v_{C,s} - \frac{1}{C_s} i_{es} \end{cases} \tag{2}$$

where i_{es} is the current of the battery. Figure 4 shows the simplified sketch of the DCMG with multiple CPLs and sources. In the Figure 5, the DC–DC boost converter is represented by a switch. u is the duty cycle of the switch and $u = 0$ shows the conducting mode and $u = 1$ shows non-conductive mode. Thus, if the dynamic equations of the DCMG system with one source and one CPL are extended, the overall dynamics model of the DCMG can be obtained. By defining the $x_i = [i_{L,i} \quad v_{C,i}]^T$ as the CPL state vector, the state-space model of the CPLs in the DCMG can be described as:

$$\begin{cases} \dot{x}_i = A_i \cdot x_i + d_i \cdot P_i + A_{is} \cdot x_s \\ y_i = x_i \end{cases} \tag{3}$$

where

$$A_i = \begin{bmatrix} -\frac{r_{L,i}}{L_i} & -\frac{1}{L_i} \\ \frac{1}{C_i} & 0 \end{bmatrix}, \; d_i = \begin{bmatrix} 0 \\ -\frac{1}{C_i v_{C,i}} \end{bmatrix}, \; A_{is} = \begin{bmatrix} 0 & \frac{1}{L_i} \\ 0 & 0 \end{bmatrix} \tag{4}$$

Similarly, the state space equations of the sources in the DCMG can be presented as:

$$\begin{cases} \dot{x}_s = A_s \cdot x_s + b_s \cdot V_{dc} + b_e \cdot i_{es} + \sum_{i=1}^{N} A_{cn} x_i \\ y_s = x_s \end{cases} \tag{5}$$

where

$$A_s = \begin{bmatrix} -\frac{r_s}{L_s} & -\frac{1}{L_s} \\ \frac{1}{C_s} & 0 \end{bmatrix}, \; b_s = \begin{bmatrix} \frac{1}{L_s} \\ 0 \end{bmatrix}, \quad A_{cn} = \begin{bmatrix} 0 & 0 \\ -\frac{1}{C_s} & 0 \end{bmatrix}, \; b_{es} = \begin{bmatrix} 0 \\ -\frac{1}{C_s} \end{bmatrix} \tag{6}$$

and $x_s = [i_{L,s} \quad v_{C,s}]^T$ is the space vector of the source. Therefore, the state space vector of the whole DCMG can be presented as:

$$\dot{X} = \bar{A}X + \bar{D}P + \bar{B}_{es} i_{es} + \bar{B}_s V_{dc} \tag{7}$$

where

$$X = [x_1^T \; x_2^T \; \ldots \; x_N^T \; x_S^T]^T, P = [P_1 \; \ldots \; P_N]^T,$$

$$\bar{A} = \begin{bmatrix} A_1 & 0 & \cdots & 0 & A_{1s} \\ 0 & A_2 & \cdots & 0 & A_{2s} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & A_N & A_{Ns} \\ A_{cn} & A_{cn} & \cdots & A_{cn} & A_s \end{bmatrix}$$

$$\bar{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & d_N \\ 0 & 0 & 0 & 0 \end{bmatrix}, \bar{B}_{es} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_{es} \end{bmatrix}, \bar{B}_s = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_s \end{bmatrix}$$

The goal of this paper is to develop an intelligent scheme to stabilize the DCMG with multiple CPLs. In the following, the non-linear droop control is proposed in detail.

3 | PROPOSED NON-LINEAR DROOP CONTROL

In a typical DCMG, several parallel DC–DC power converters are connected to the DC bus, which injects power to the CPLs. It is obvious that in the DCMG the power rating of each converter is different.

Based on the power rating of each converter, the connecting cables have different equivalent resistance which affects

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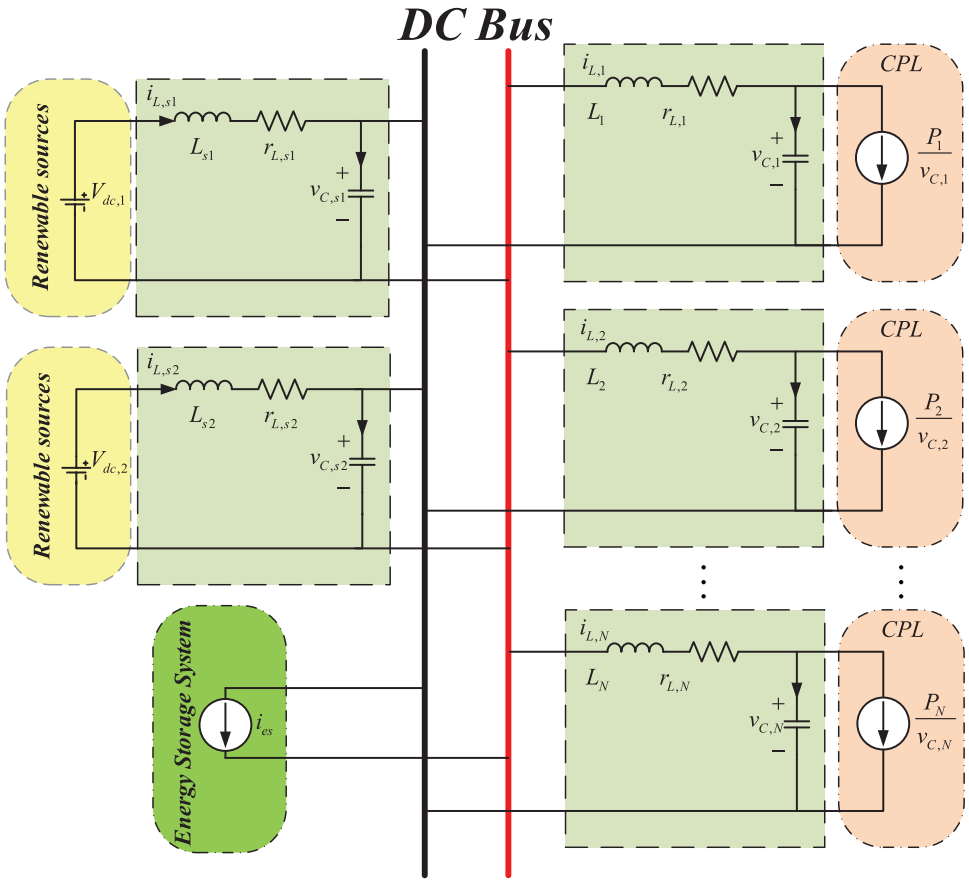


FIGURE 4 Modelling of the DCMG with N CPLs

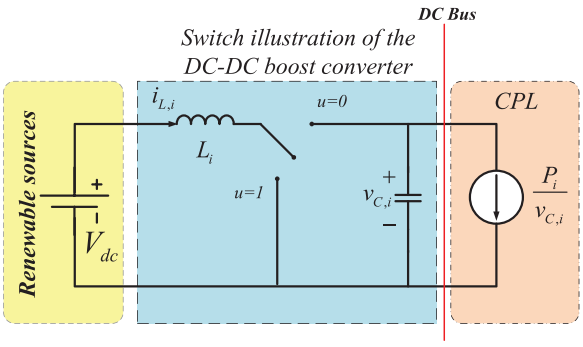


FIGURE 5 Switch illustration of the DC–DC boost converter

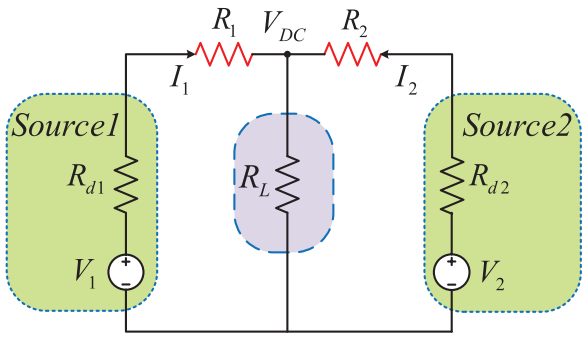


FIGURE 6 Modelling of the DCMG with N CPLs

the current sharing among paralleled power converters. It means that some converters meet the over-current situation, maybe can damage the converter's switches. To show this, an equivalent circuit consisting of two parallel DC–DC power converters is considered, which are loaded by CPLs.

As depicted in Figure 6, V_1 and V_2 are the voltage, I_1 and I_2 are the current of each converter, respectively. R_1 and R_2 are the line resistances, R_{d1} and R_{d2} are the droop resistance of the converters which can be defined as follows:

$$R_{di} = \frac{\Delta V}{I_{\max}} \tag{8}$$

where $I_{\max}$ is the maximum injected current of the power converter and ΔV is the allowable voltage error from the nominal value. Therefore, the total equivalent resistance from the converters to the CPLs can be obtained as:

$$R'_{d1} = R_{d1} + R_1 \tag{9}$$

$$R'_{d2} = R_{d2} + R_2 \tag{10}$$

Based on Figure 4, the current difference of parallel converters can be calculated as follows:

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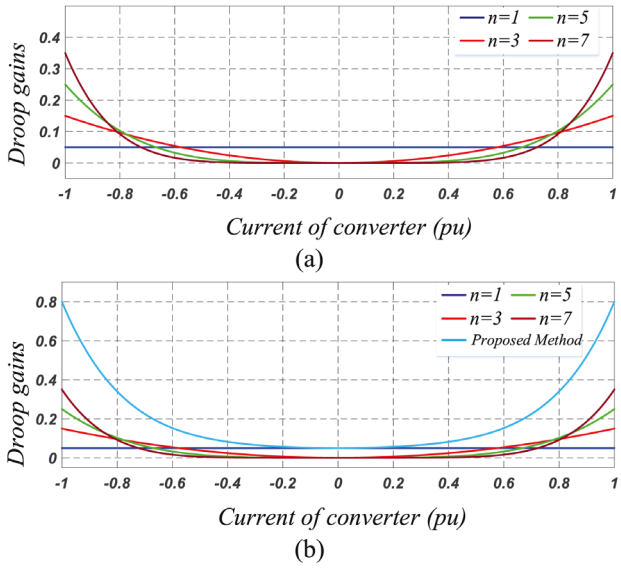


FIGURE 7 Proposed non-linear droop control (a) droop gains for different values of *n*. (b) Droop gain for the proposed method for $n = 1,3,5,7$

$$I_1 - I_2 = \frac{2(V_1 - V_2)}{(R'_{d1}) + (R'_{d2}) + \frac{(R'_{d1})\cdot(R'_{d2})}{R_L}} + \frac{V_1 \cdot (R'_{d2}) - V_2 \cdot (R'_{d1})}{(R'_{d1}) \cdot (R'_{d2}) + (R'_{d1}) \cdot R_L + (R'_{d2}) \cdot R_L} \quad (11)$$

It can be concluded that the larger droop gain provides better current sharing among parallel converters, but it can increase the voltage deviation in the DCMG [23]. The main reason for this fact is that high droop gain leads to improving current sharing while high droop gain affects the voltage deviation. Therefore, the designer should set a trade-off between the voltage regulation and current sharing in the design procedure of the droop control method.

In order to have proper current sharing among parallel converters in the DCMG by keeping voltage deviation in the acceptable range, the optimization of the droop curve is required. The general droop control technique can be formulated as:

$$V_i = V^* - R_{di} \cdot \left(I_i^n\right), n \in R^+ \quad (12)$$

where R_{di} is the virtual resistance, V_i is the voltage of the *i*th power converter, V^* is the reference voltage of the DCMG, and I_i is the current of the *i*th power converter. For a specific voltage deviation, various droop curves can be obtained with different values of n. Figure 7 presents the droop curves based on the output current of the bidirectional DC–DC converter. When $i < 0$ the power converter is sinking the power and for $i > 0$ the power converter is outputting the power. For $n > 1$, the droop gains increase when the output current is increased, and for $n = 1$, the droop gain is constant irrespective of the output current of the bidirectional converter, that is, linear droop method. It should be noted that the values of n are selected from odd numbers to have symmetrical droop curves. As shown in Figure 7, it can be seen that in the heavy load condition the linear droop scheme has a small droop gain, whereas the nonlinear droop method has a small droop gain in the light loading condition. Thus, in order to achieve optimum droop gain in all loading conditions for desired current sharing, a novel non-linear droop scheme is presented here.

To achieve a fast dynamic response in the power converters, a single current controller is replaced with the dual inner current controller and outer voltage controller, which is named *I–V* droop controller:

$$I_{ref,i} = \frac{v^* - v_i}{R_{di}} \quad (13)$$

where $I_{ref,i}$ is the reference current, and v_i is the voltage measured from the *i*th converter. Considering Equation (12) the origin *I–V* droop control can be expressed as:

$$v_i = v^* - R_{di,1} \cdot I_i - R_{di,n} \cdot I_i^n \quad (14)$$

Finally, by substituting Equation (14) into Equation (13) the proposed nonlinear droop control method can be formulated as:

$$I_{ref,i} = \frac{v^* - v_i}{R_{di,1}} - \left(\frac{R_{di,n}}{R_{di,1}} \cdot I_i^n\right) \quad (15)$$

The values of n are adopted depending on the maximum current of the *i*th converter and voltage deviation. As depicted in Figure 7, it can be concluded that the suggested scheme has the maximum available droop gain in various loading conditions from light-load to full-load conditions considering the line resistance in each converter. Therefore, the desired current sharing among parallel power converters can be realized with the suggested controller.

4 | PROPOSED MODEL-FREE CONTROLLER

In this section the design procedure of a model-free control (MFC) scheme involves of three subsystems is presented to improve the stability of parallel power converters in the DCMG loaded by CPLs. The overall diagram of the proposed controller is shown in Figure 8. First, in order to estimate the value of the unmodelled dynamics of the CPLs, an ESO is employed. Second, MFC-based ISIT2-FLC is presented to provide control requirement, high order derivative output suppression, and decrease the difficulty of the present controller. Lastly, an SMC has been used to eliminate the error of the ESO.

4.1 | The SIT2-FLC based PI controller

4.1.1 | Structure of the IT2-FPI

Figure 9a depicts the overall structure of the SIT2-FPI controller. The scaling factor G_e is adopted to normalize the error signal *e*. In this structure, the error signal, after passing the input

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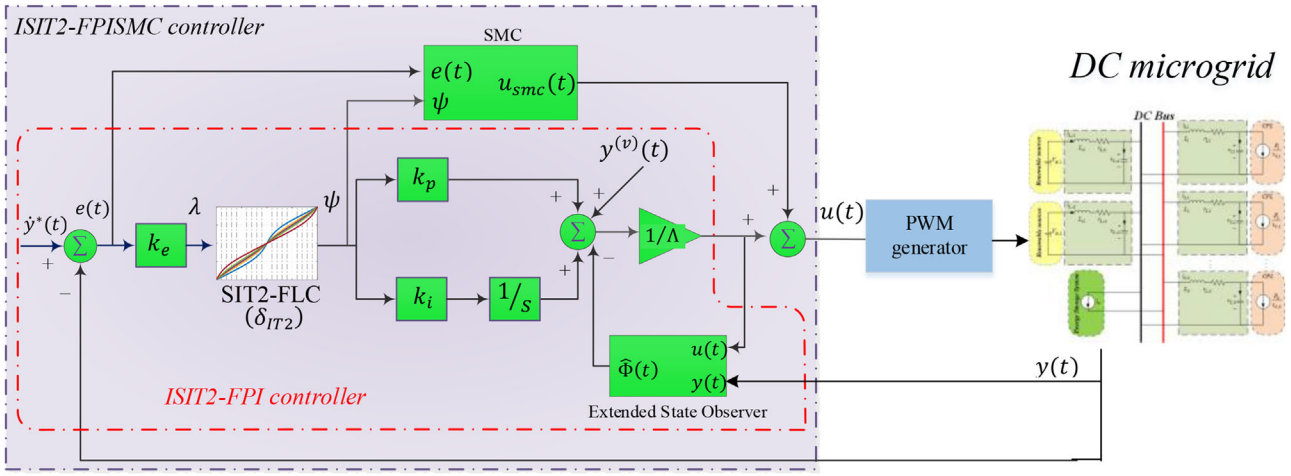


FIGURE 8 The block diagram of the proposed controller

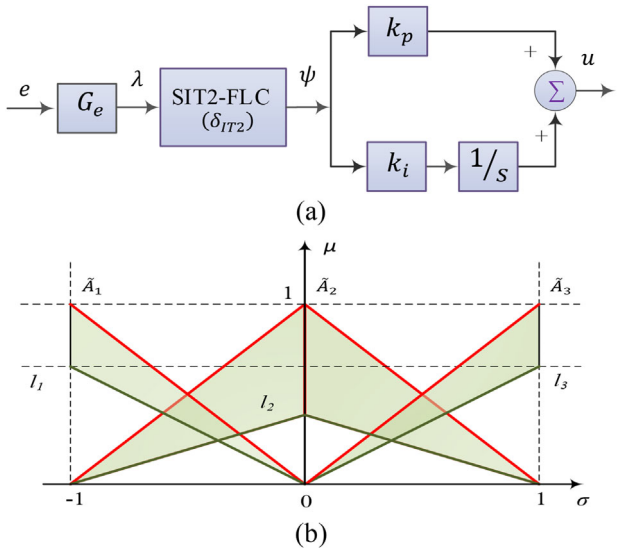


FIGURE 9 Structure of (a) SIT2-FPI controller. (b) Triangular member functions

scaling factor, is converted into λ, which is the input of interval type 2. The control signal (u) is generated by the output of the SIT2-FPI controller (ψ), given as follows:

$$u(t) = k_p\psi + k_i \int \psi \tag{16}$$

where

$$k_p = k_{p0}G_u; k_i = k_{i0}G_u \tag{17}$$

As depicted in Figure 9a, $k_{p0}kp0$ and k_{i0} present the PI controller constants and G_u represent the output scaling factor, which is defined as $G_u = G_e^{-1}$.

4.1.2 | Design of SIT2-FLC

For a SIT2-FLC, the rule structure is described as [24]:

$$r_i : \text{IF } e \text{ is } \tilde{A}_i \text{ THEN } e_{IT2} \text{ is } B_i, \; i = 1, 2, 3 \tag{18}$$

As shown in Figure 9b, triangular types of IT2 fuzzy sets $\tilde{A}_i$ made the membership functions (MFs). B_i represent the crisp consequents values and defined as $B_1 = -1$, $B_2 = 0$ and $B_3 = 1$. Moreover, the lower MF ($\underline{\mu}_{\tilde{A}_n}$) and upper MF ($\bar{\mu}_{\tilde{A}_n}$) are used to define the IT2FSs. The height of the lower MFs is defined as l_i, where $i = 1$, 2, and 3. Here, with $l_1 = l_3 = 1 - \delta_{IT2}$ and $l_2 = \delta_{IT2}$, the symmetrical MFs are used. δ_{IT2} is the main controller coefficient which is adopted based on the desired performance of the plant.

Considering the centre of the type reduced set, the defuzzified output of the IT2-FLC can be expressed as follows [25]:

$$\varphi_o = \left(\varphi_o^r + \varphi_o^l\right)/2 \tag{19}$$

where the φ_o^r and φ_o^l are the endpoints of the set type reduction scheme, which are expressed as:

$$\varphi_o^l = \frac{\sum_{n=1}^{L} \bar{\mu}_{\tilde{A}_n}(\sigma) \cdot B_n + \sum_{n=L+1}^{N} \underline{\mu}_{\tilde{A}_n}(\sigma) \cdot B_n}{\sum_{n=1}^{L} \bar{\mu}_{\tilde{A}_n}(\sigma) + \sum_{n=L+1}^{N} \underline{\mu}_{\tilde{A}_n}(\sigma)} \tag{20}$$

$$\varphi_o^r = \frac{\sum_{n=1}^{R} \underline{\mu}_{\tilde{A}_n}(\sigma) \cdot B_n + \sum_{n=R+1}^{N} \bar{\mu}_{\tilde{A}_n}(\sigma) \cdot B_n}{\sum_{n=1}^{R} \underline{\mu}_{\tilde{A}_n}(\sigma) + \sum_{n=R+1}^{N} \bar{\mu}_{\tilde{A}_n}(\sigma)} \tag{21}$$

In Equations (20) and (21), R and L denote the points of switching.

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TABLE 1 Specification of ISIT2-FLC under various values of δ_{IT2}

	Design variables	Type of CC
Mode-1	If $0 < \delta_{IT2} < \delta_{\epsilon 1}$, then $\varepsilon > 0$ for $\forall\sigma\varepsilon(0, 1]$ and $\delta_{\epsilon 1} = (3-\sqrt{5})/2$	Aggressive CC_{IT2} (A-CC_{IT2})
Mode-2	If $\delta_{\epsilon 2} < \delta_{IT2} < 1$, then $\varepsilon < 0$ for $\forall\sigma\varepsilon(0, 1]$ and $\delta_{\epsilon 2} = (\sqrt{5}-1)/2$	Smooth CC_{IT2} (S-CC_{IT2})
Mode-3	If $\delta_{\epsilon 1} < \delta_{IT2} < \delta_{\epsilon 2}$, then $\varepsilon < 0$ for $\forall\sigma\varepsilon[0.5, 1]$ and $\delta_{\epsilon 1} = (3-\sqrt{5})/2$	Moderate CC_{IT2} (M-CC_{IT2})

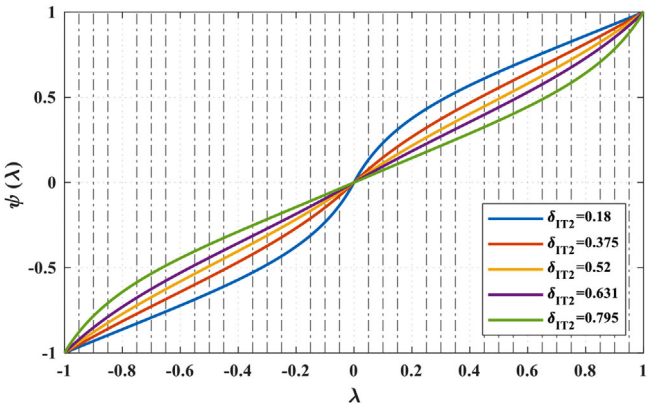


FIGURE 10 Control surfaces based on different values of δ_{IT2}

Also, in the SIT2-FLC, the fuzzy mappings (FM) can be represented as [26]:

$$\psi(\sigma) = \sigma \cdot k(|\sigma|) \tag{22}$$

where $k(\sigma)$ is a nonlinear term, which can be expressed as:

$$k(e) = \frac{1}{2}\left(\frac{1}{\delta_{IT2} + \lambda - \delta_{IT2}\lambda} + \frac{\delta_{IT2} - 1}{\delta_{IT2}\lambda - 1}\right) \tag{23}$$

For various constraints of δ_{IT2}, by defining $\varepsilon_o(\lambda) = \psi(\lambda) - \lambda$, the three mapping IT2-FLC modes based on control curve (CC) can be achieved, as illustrated in Table 1. As shown in Figure 10, examples of A-CC_{IT2}, S-CC_{IT2} and M-CC_{IT2} are presented for $\delta_{IT2} = 0.18$, $\delta_{IT2} = 0.375$, $\delta_{IT2} = 0.52$, $\delta_{IT2} = 0.631$ and $\delta_{IT2} = 0.795$. More information about the control curves generations and design of the SIT2-FLC can be found in ref. [25].

4.2 | ISIT2-FLC based SMC

4.2.1 | Intelligent SIT2-FLC based PI controller

In a typical non-linear system with single input and single output, considering an ultra-local model, the model-free method can be expressed as [27]:

$$y^{(v)}(t) = \Phi + \Lambda u(t) \tag{24}$$

where $y^{(v)}(t)$ is usually chosen as 1 or 2 and is the derivative of the output $y(t)$ with order $v \in \mathbb{N}$. In order to have the same order of magnitude for $\Lambda u(t)$ and $y^{(v)}(t)$, $\Lambda \in \mathbb{R}$ can be defined as a nonphysical design constant. Also, Φ denotes an unknown term and recognize by the output $y(t)$ and control signal(t).

By considering Φ and Λ as well-known parameters, an ISIT2-FPI can be characterized as follows:

$$u(t) = \frac{1}{\Lambda}\left(-\hat{\Phi} + y^{(v)^*} + k_p\psi_u + k_i\int\psi_u\right) \tag{25}$$

In Equation (25), $y^{(v)^*}$ is the desired reference. Moreover, $\hat{\Phi}$ is the estimated of $\Phi\Phi t$. By combining Equations (24) and (25), the error of the output, that is, $e(t) = y^*(t) - y(t)$ can be achieved as follows:

$$e^{(v)} + \hat{\Phi} + k_{ps}\psi_u + k_{is}\int\psi_u = 0 \tag{26}$$

Remark: The intelligent controllers in the framework of model-free approaches need an observer based on time delay estimation or algebraic techniques to estimate the unmodeled dynamics and uncertainties. However, these observers are not able to guarantee an accurate tracking error in the presence of uncertainties. Therefore, an extended state observer (ESO) is integrated into an intelligent single input interval type-II fuzzy logic controller (ISIT2-FLC) to estimate the unknown uncertain dynamics of the system.

Here, by using the ESO, the estimated disturbance $\hat{\Phi}$ will be calculated. The ESO is designed based on a third observer as [28]:

$$\begin{aligned}
\dot{q}_1 &= q_2 - \mu_{o1}\, fal\,(F, \omega_1, \rho)\\
\dot{q}_2 &= q_3 - \mu_{o2}\, fal\,(F, \omega_2, \rho)\\
\dot{q}_3 &= -\mu_{o3}\, fal\,(F, \omega_3, \rho)\\
fal\,(F, \omega_1, \rho) &= \begin{cases} |F|^{\omega}\mathrm{sgn}(F), & |F| > \rho \\ F/\rho^{1-\omega}, & otherwise \end{cases} \quad 0 \le \omega \le 1,\ \rho > 0
\end{aligned} \tag{27}$$

where F denotes the error estimation of the observer ($F = y - z_1$), q_1, q_2 and q_3 represent the intermediate states, and μ_{o1}, μ_{o2} and μ_{o3} are the observer constant. In the ESO, q_3 shows the estimation of the observer, therefore $q_3 = \hat{\Phi}$.

4.2.2 | Model-free sliding mode control

From Equation (25), it can be concluded that the effectiveness of the proposed controller depends on three items; (1) the coefficient of the PI controller in the SIT2-FLC; (2) estimation of Φ; and (3) the Footprint of Uncertainty (FOU) design coefficient. The inaccurate estimation can degrade the performance of the proposed controller. Therefore, in order to decrease the estimation error, an SMC is employed. Figure 8 shows the overall diagram of the ISIT2-FPI with the proposed SMC. In Figure 8,

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the term $y(t)$ is the output voltage and $y^{(v)^*}$ denotes the reference voltage with the order v.

Let u_{smc} as an additional SMC input control, then:

$$u(t) = \frac{1}{\Lambda}\left(-\hat{\Phi} + y^{(v)^*} + k_p\psi_u + k_i\int\psi_u\right) + u_{smc} \quad (28)$$

Also, $\tilde{\Phi} = \hat{\Phi} - \Phi$ is defined as the estimated error of the observer. Here $\|\tilde{\Phi}\| < \Phi_m$ which Φ_m is an upper bound. By selecting $v = 2$ and replacing Equations (28) into (24), the equation of the error can be obtained as:

$$\ddot{e}(t) + \Lambda u_{smc} + \tilde{\Phi} + k_p\psi_u + k_i\int\psi_u = 0 \quad (29)$$

A switching function in the SMC design scheme is presented as:

$$\sigma = \dot{e} + \gamma e \quad (30)$$

In Equation (30) $\gamma > 0$ is a design term. The derivative of Equation (30) can be written as:

$$\dot{\sigma}(t) = \gamma\dot{e} + \ddot{e} = \gamma\dot{e} - k_p\psi_u - k_i\int\psi_u - \Lambda u_{smc} - \tilde{\Phi}(t) \quad (31)$$

The extra input u_{smc} includes the switching control (u_{sw}) and the equivalent control (u_{eq}). In order to realize the proper SMC condition in Equation (31), u_{eq} is achieved by substituting $\tilde{\Phi}$ with Φ_m as follows:

$$u_{eq} = \left(\gamma\dot{e} - \Phi_m - k_p\psi_u - k_i\int\psi_u\right)/\Lambda \quad (32)$$

Moreover, to remove the chattering effects, u_{sw} is determined as:

$$u_{sw} = (\eta_1.[sat(\sigma,\varepsilon)] + \eta_2.\sigma)/\Lambda \quad (33)$$

where η_1 and η_2 represent the switching gains which play critical role in the performance of SMC. Also, ε represents the boundary thickness, $\eta_1 > 0, \eta_2 > 0, \varepsilon > 0$ and:

$$sat(\sigma,\varepsilon) = \begin{cases} 1 & if\ \sigma > \varepsilon \\ \sigma/\varepsilon & if\ \|\sigma\| > \varepsilon \\ -1 & if\ \sigma < -\varepsilon \end{cases} \quad (34)$$

Substituting u_{smc} into Equation (28), the control signal of the proposed controller can be obtained as:

$$u(t) = \left(\eta_1.[sat(\sigma,\varepsilon)] + \eta_2.\sigma + \gamma.\dot{e} - \Phi_m - \hat{\Phi} + y^{(v)^*}\right)/\Lambda \quad (35)$$

4.2.3 | Stability evaluation

In order to prove the stability of the controller, a Lyapunov function can be considered as:

$$L = \frac{1}{2}\sigma^2 \quad (36)$$

The derivation of Equation (36) is calculated as follows:

$$\dot{L} = \sigma.\dot{\sigma} = -\sigma\left(\tilde{\Phi} - \Phi_m + \eta_1[sat(\sigma,\varepsilon)]\right) - \eta_2.\sigma^2 \quad (37)$$

To show that $\dot{L} < 0$, it is enough to show that $\tilde{\Phi}$ is boundedness. The stability of the defined Lyapunov function can be guaranteed under two conditions as follows [29, 30]:

State1: If $\varepsilon < |\sigma|$, the situation will be transformed as:

$$\dot{L} = -\sigma\left(\tilde{\Phi} - \Phi_m + \eta_1\right) - \eta_2.\sigma^2 \quad (38)$$

If one has $\eta_1 > 2\Phi_m$, since $\tilde{\Phi}$ is boundedness, it is meet that $\dot{L} < 0$

State2: If $\varepsilon > |\sigma|$, the situation will be converted as:

$$\dot{L} = -\sigma\left(\tilde{\Phi} - \Phi_m + \eta_1.\sigma/\varepsilon\right) - \eta_2.\sigma^2 \quad (39)$$

Based on the condition of Equation (39), it is met that the term right-side is negative.

$$\begin{aligned} -(\eta_1/\varepsilon + \eta_2)\sigma^2 - \sigma\left(\tilde{\Phi} - \Phi_m\right) < 0 \\ (\eta_1/\varepsilon + \eta_2)\sigma > \left|\tilde{\Phi} - \Phi_m\right| \end{aligned} \quad (40)$$

According to the boundedness of $\tilde{\Phi}$ and considering $|\sigma| < \varepsilon$, it is concluded that $2\Phi_m < (\eta_1 + \varepsilon\eta_2)$. In the following section, to confirm the proper performance of the proposed scheme and the stability of the DCMG with several CPLs, the experimental results are presented.

Remark: The studied DCMG depicted in Figure 4 is made from several subsystems with three DGs to supply the CPLs. In this application, three distinct controllers are established to regulate the switches of each converter. Therefore, three input signals of controller-1 ($u_1(t)$), controller-2 ($u_2(t)$), and controller-3 ($u_3(t)$) are applied to the DG1, DG2, and DG3, respectively.

5 | ADVANTAGES OF THE PROPOSED CONTROL TECHNIQUE

The present work focuses on the deployment of interval type-II fuzzy logic controllers in a model-free framework to improve the efficiency of DC/DC power converters. Many considerations have been adopted in the proposed control methodology which offers significant benefits from a practical perspective.

1. As some linear control approaches such as linear quadratic regulators (LQR) are adopted for power converters, they will lose some features of the system when linearizing the

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TABLE 2 Specification of the plant

	Parameters	value
Electrical parameters	Voltage of DC bus	400
	Converter inductance	1800 μH
	Converter capacitance	2200 μF
	Line resistance for DG1	0.3 Ω
	Line resistance for DG2	0.6 Ω
	Line resistance for DG3	0.9 Ω
Droop control parameters	k_p	3
	k_i	15
	k_1	0.2

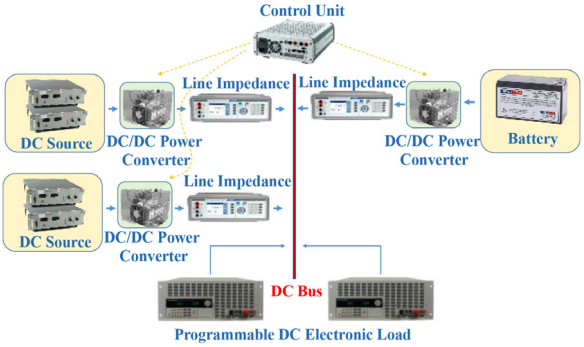


FIGURE 11 Experimental setup

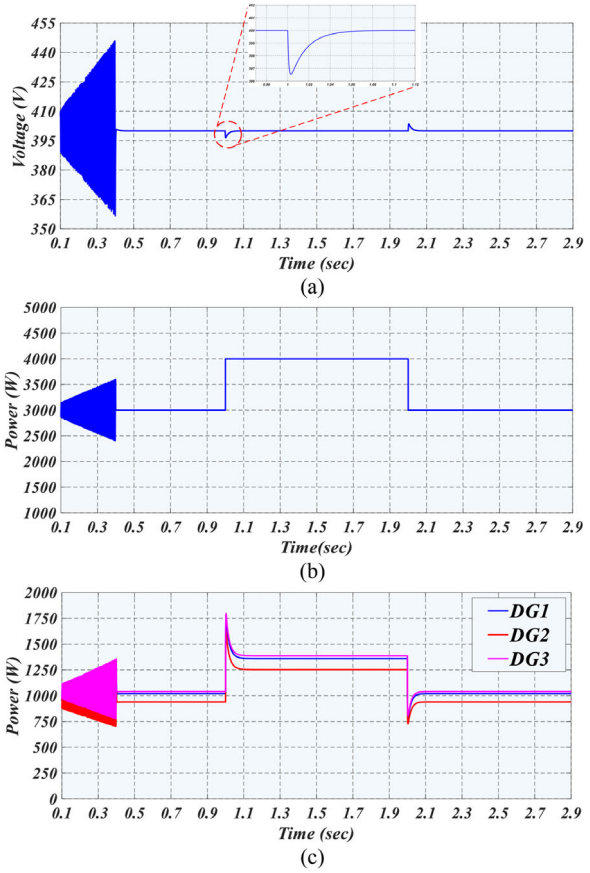


FIGURE 12 Experimental outcomes with the suggested controller and conventional droop control; (a) voltage of DC bus, (b) power of the CPLs, and (c) power of the parallel converters

dynamic model. However, the shortcomings of linear controllers have been addressed by adopting the proposed non-linear controller.

2. The robust model-based controllers such as H-infinity and MPC need to model the system in their process design which limits their applicability in real-world systems. Unlike the robust controllers, the proposed controller is designed based on the input and output data of the system in a model-free approach.
3. The light burden of computations is another benefit of the proposed controller, which has a prominent role in the practical development and online control test systems.
4. Finally, to evaluate the usefulness and efficiency of the proposed model-free controller, many real-time examinations under various conditions of CPL's power have been made.

6 | HIL EXPERIMENTAL RESULTS

To validate the efficacy of the proposed controller, a DC Microgrid with three parallel converters is considered. Table 2 shows the parameters of the plant. In order to exploit the properties of both A-CCIT2 and S-CCIT2, the value of δ_{IT2} is set to 0.52 (moderate CCIT2). The experimental setup of the studied pawer electronic test system (described in Section 2) is shown in Figure 11. In this setup, a dSPACE MicroLabBox with DS1202 Power PC Dual Core 2 GHz processor. The MicroLabBox platform with the real-time interface (RTI) is adopted for the design procedure.

The voltage of the DCMG is 400 V and the nominal output power of each converter are considered to be 1.5 kW. Figure 12a shows that the voltage of the DCMG is oscillating, which confirms the instability effect of the CPL. In fact, when the input and output powers of the LC are not equal in a steady-state, the energy is an imbalance, which is caused by the oscillation in the voltage. Therefore, it is necessary to enhance the stability of the system with CPL. As depicted in Figure 12a, before $t = 0.4$ s the proposed controller is not active and the DC bus voltage of the DCMG and the power of the paralleled converters are oscillating. From $t = 0.4$ s the proposed controller is adopted and consequently, the system is stable.

In order to show the performance of the suggested controller in a dynamic situation, a step load change is applied in the DC bus of the DCMG. Figure 12b shows that the power of the load is changed from 3 to 4 kW at $t = 1$ s and restored at $t = 2$ s. From Figure 12c it can be concluded that during this step load changes the system is stable and these curves indicate a good transient response. Figure 12c shows the performance of the DCMG with a traditional droop control scheme. Since the traditional droop control method is adopted for power-sharing among paralleled converters in the DCMG, it cannot

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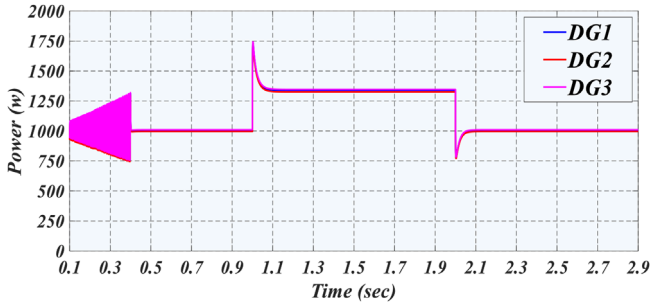


FIGURE 13 Power of the parallel converters with the ISIT2-FLC and the suggested nonlinear droop control during step change load

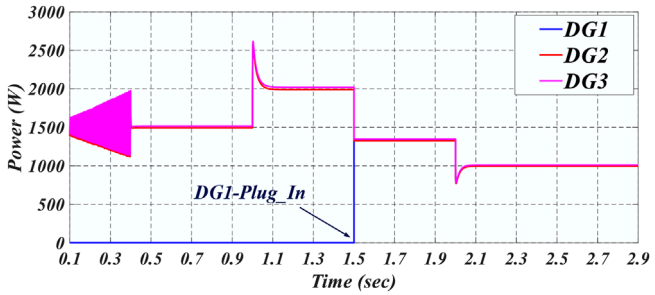


FIGURE 14 Power of the converters for plug-and-play experimental results with the proposed controller and the proposed nonlinear droop control

provide proper power-sharing among paralleled converters, which is maybe caused by the current situation in the DC–DC converters. To improve the current sharing in the DCMG, the proposed nonlinear droop control is used.

As illustrated in Figure 13 when the DCMG operates with the proposed nonlinear droop control approach, the accuracy of the power-sharing among paralleled converters is enhanced compared with Figure 12c. To show the Plug-and-Play capability of the proposed controller in the DCMG, Figure 14 illustrates the power of the DGs during this condition. As depicted in Figure 14, before $t = 1.5$ s DG2 and DG3 are injected power to the load. At $t = 1.5$ s the DG1 is connected to the DCMG and starts the power injection to the CPL. From Figure 14 it can be concluded that the proposed nonlinear droop control method can guarantee accurate power-sharing among DGs during this situation and provides plug-and-play capability for DGs in the DCMG. Moreover, in order to show the effectiveness of the suggested non-linear droop control method with ISIT2-FLC against other controllers, a step load change is applied in the DCMG. The comparison of the suggested controller with the MPC controller and PI controller is drawn in Figure 15. From Figure 15 it can be observed that the proposed control scheme has faster and proper transient outcomes during step load change.

7 | CONCLUSION

Here, a new non-linear droop control method was proposed for paralleled converters in the DCMG loaded by CPLs. The

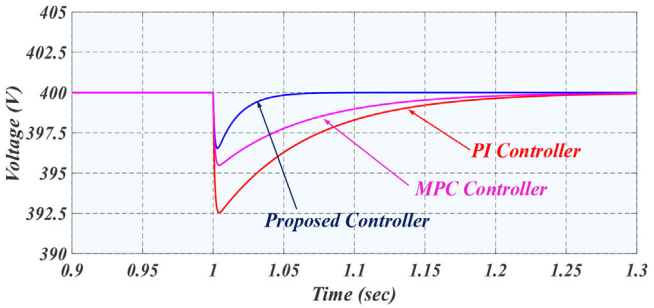


FIGURE 15 Comparison of the transient outcomes of the ISIT2-FLC, MPC and PI controller

proposed control method improves the power-sharing among paralleled converters during dynamic and steady-state situations. Moreover, a novel model-free controller was introduced to guarantee the stability of the DCMG. A single input interval type-II fuzzy logic controller (ISIT2-FLC) was developed to improve the stability of the converters feeding the CPLs. The proposed controller estimates the unknown dynamics in the DC–DC converters loaded by CPLs with an extended state observer (ESO). Furthermore, a sliding mode control (SMC) method was proposed to eliminate the error estimation of the ESO. The HIL experimental examinations validated that with the application of the suggested model-free controller, the quality of voltage waveforms in the terms of overshoot and settling time have been improved than the conventional controllers. Thus, a higher level of stability of the DCMG is realized by the proposed controller during different conditions of CPL's power.

CONFLICT OF INTEREST

The authors do not have a conflict of interest.

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